ECA08 – ANALOG AND DIGITAL COMMUNICATION

CO3 Application-Oriented Assignment – Engineering Design Report

Digital Voice and Data Link Design: From Sampling and Pulse Transmission to Deployment

Name: P. Harish Register No.: 1913260080 Date: 6 November 2026

Flow of the report: REQUIREMENT → ALTERNATIVES → ANALYSIS → SELECTION → DESIGN → REALIZATION → VERIFICATION → DEPLOYMENT DECISION

Summary of the design (details and proofs follow)

Sampling and coding: f_s = 8 kHz (minimum 6.8 kHz), anti-alias filter 3.4 kHz (elliptic, order 5), 8-bit μ-law PCM (G.711 type), 64 kb/s per voice channel; uniform 8-bit SQNR 49.9 dB for a full-scale sine; μ-law gives about 37 dB over a 40 dB level range.

Pulse modulation for control: PWM (best match to motor/actuator drive). **Pulse shaping:** raised cosine, α = 0.35, with MMSE equalisation; MLSE for severe ISI. **Multiplexing:** TDM, 105-bit frame every 125 μs, 840 kb/s aggregate.

Modulation: QPSK, E_b/N_0 = 9.6 dB for BER 10^{-5}, bandwidth 567 kHz (RC α = 0.35); coherent detection with Costas carrier recovery and Gardner timing recovery. **Decision:** YES, deploy (VoIP, DSL and fibre mappings given in Stage 7).

Stage 1 – Interpret the Digital Communication Design Requirement

1.1 End-to-end digital communication block diagram

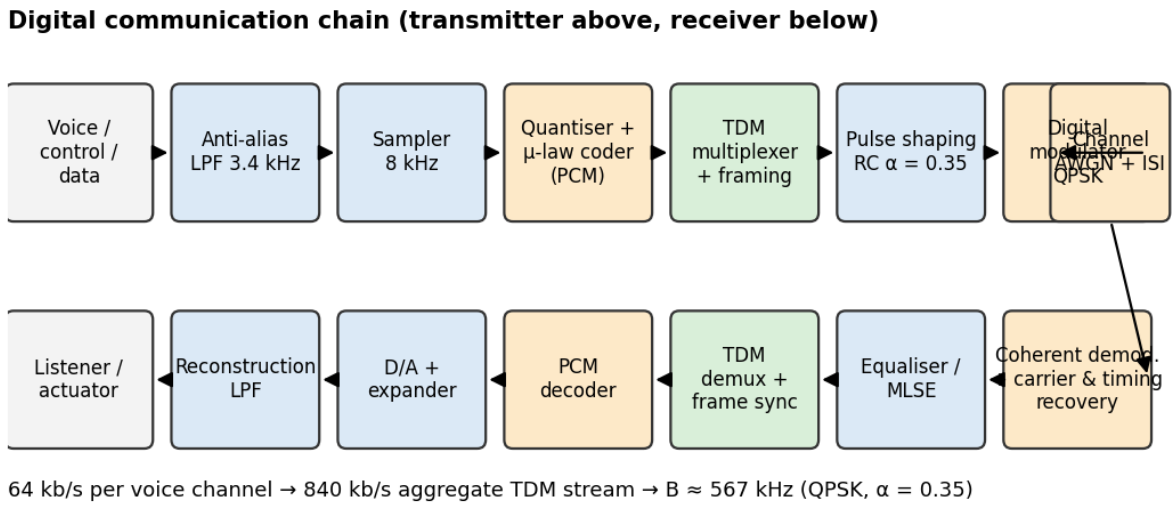


Fig. 1 Complete end-to-end digital communication system (transmitter above, receiver below)

The voice (3.4 kHz) is band-limited by an **anti-alias filter**, **sampled**, **quantised** and **encoded** (PCM) into 64 kb/s. The control and monitoring streams are combined with the voice in a **TDM multiplexer** that inserts a framing bit. The bit stream is shaped by a **raised-cosine pulse-shaping filter** and **QPSK-modulated** onto the line or radio carrier. The **channel** is band-limited and adds AWGN and ISI. The receiver performs **coherent demodulation with carrier and timing recovery**, **equalisation/MLSE**, **frame synchronisation and demultiplexing**, **PCM decoding and expansion**, and **reconstruction** (low-pass filtering) to deliver voice or control data.

1.2 Minimum sampling frequency and suitability of 8 kHz

By the Nyquist criterion $f_s \geq 2 f_m = 2 \times 3.4 =$ **6.8 kHz**. The baseline 8 kHz leaves a guard band $8 - 2\times3.4 =$ **1.2 kHz** (the anti-alias filter transition from 3.4 kHz to 4.6 kHz = $f_s - 3.4$ kHz). 8 kHz is therefore **suitable** and is also the standard telephony rate (125 µs frame). Its drawback is that a very sharp analogue filter is needed: for 1 dB pass-band ripple and 40 dB stop-band attenuation a Butterworth filter would need order 18, an elliptic filter only order 5 (Fig. 2c). **Alternative if required:** sample at 32 kHz (4× oversampling) with a simple analogue filter (Butterworth order 3, because aliases now start at 28.6 kHz) followed by digital decimation to 8 kHz; this is the choice for integrated codecs. For wider-band voice (50–7000 Hz) 16 kHz would be required, which is not needed for industrial commands.

1.3 Anti-aliasing and demonstration of under-sampling

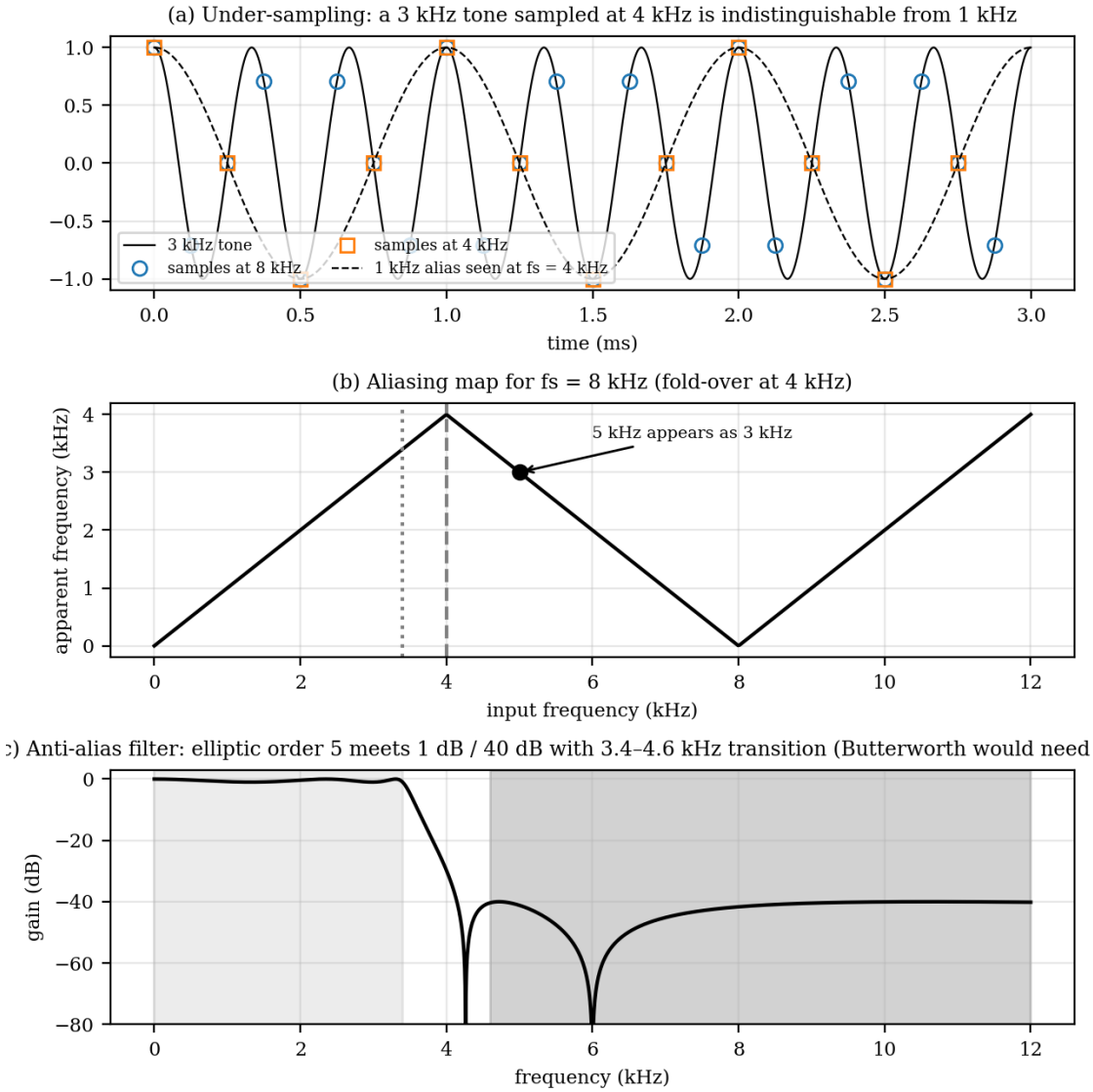


Fig. 2 Under-sampling, aliasing map and anti-alias filter (simulation)

Interpretation: in panel (a) a 3 kHz tone sampled at 4 kHz (below its Nyquist rate of 6 kHz) produces samples identical to those of a 1 kHz tone: the information is irreversibly corrupted, because $f_{alias} = |f - k f_s| = |3 - 4| = 1$ kHz. Panel (b) shows that with $f_s = 8$ kHz a 5 kHz component folds to 3 kHz, inside the voice band, which is why every component above $f_s - 3.4$ kHz = 4.6 kHz must be removed before sampling (panel c).

1.4 Basic information rate

$R_b = f_s \times n = 8\,000 \times 8 =$ **64 kb/s** per voice channel. The minimum (Nyquist) transmission bandwidth for 64 kb/s is $R_b/2 = 32$ kHz; with a raised-cosine pulse ($\alpha = 0.35$) the baseband bandwidth is $(1+\alpha)R_b/2 =$ **43.2 kHz**.

1.5 Constraints

- **Bandwidth:** band-limited channel (twisted pair or radio) forces pulse shaping and limits symbol rate; **noise:** AWGN sets the E_b/N_0 needed (Stage 6); **ISI:** dispersion of the channel; **timing:** clock and frame synchronisation; **implementation:** low-cost codec, fixed-point processing, 125 μs frame.

1.6 VoIP, DSL and fibre-optic requirements

Application	Dominant requirement	Typical differences
VoIP	Delay, jitter and packet loss; interoperability (G.711/G.726)	Voice packetised in RTP/UDP/IP; 64 kb/s payload becomes 80 kb/s on the IP layer; low-rate coding is attractive to save bandwidth
DSL	Bandwidth-limited twisted pair, crosstalk, strong ISI	Needs equalisation, echo cancellation and bit loading; DMT/QAM modulation, timing recovery
Fibre optic	Very high bit rate, dispersion, clock recovery	Low noise; NRZ/PAM on intensity, TDM (SDH/OTN) plus WDM; clock-data recovery

Stage 1 deliverable – Digital Communication Design Requirement Statement

Sampling: 8 kHz (≥ 6.8 kHz), elliptic anti-alias filter 3.4–4.6 kHz. **Quantisation:** 8 bits (256 levels), μ-law companding. **Initial bit rate:** 64 kb/s per voice channel; 840 kb/s multiplexed.

Channel: band-limited, AWGN and ISI. **Applications:** industrial voice/control, VoIP, DSL, fibre. **Objectives:** SQNR ≥ 35 dB over a 40 dB range, BER ≤ 10^{-5} at the operating E_b/N_0, zero-ISI shaping, frame synchronisation with < 1 μs jitter.

Stage 2 – Quantisation and Voice-Coding Technique

2.1 8-bit uniform quantiser

Quantity	Calculated	Simulated
Number of levels $L = 2^n$	256	256
Step size $\Delta = 2V_{max}/L$ ($V_{max} = 1$ V)	7.813 mV	7.813 mV
Maximum quantisation error	$\pm\Delta/2 = \pm 3.906$ mV	±3.906 mV (uniform histogram)
Error power $\Delta^2/12$	5.086×10^{-6} V²	5.049×10^{-6} V²
SQNR, full-scale sine: 6.02n + 1.76	49.9 dB	49.9 dB

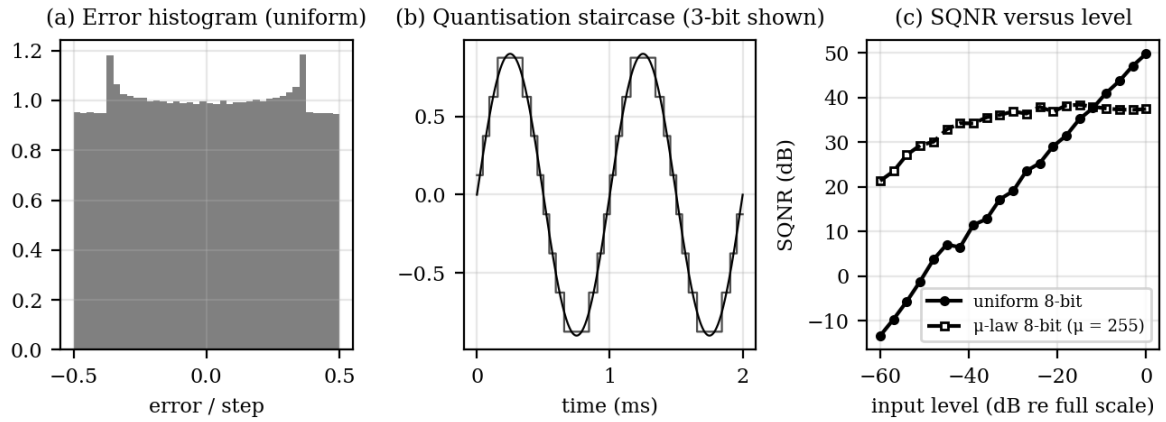


Fig. 3 Quantisation error, staircase and SQNR versus level for uniform and μ-law 8-bit quantisers

Interpretation: theory and simulation agree within 0.03 dB for the full-scale sine. The uniform quantiser has constant noise, so SQNR falls 1 dB for every 1 dB of level reduction (49.8 dB at full scale, 29.0 dB at −21 dB and only 11.5 dB at −39 dB), which is a problem for speech whose level varies by 40 dB or more. The μ-law quantiser (μ = 255) compresses the signal before the uniform quantiser and keeps SQNR between 34.2 and 37.9 dB over the same range.

2.2 Comparison of PCM, DPCM, DM, ADM and ADPCM

All coders were run on the same 2 s speech-like test signal (formant-filtered pulse train, 300–3400 Hz, syllabic level variation of about 18 dB, rms = -20.0 dBFS). DM/ADM operate at oversampled rates (64 kHz and 32 kHz). SNR is measured over the whole signal and as the segmental SNR over 20 ms frames.

Coder	Rate (kb/s)	SNR (dB)	Segmental SNR (dB)	Bandwidth (min. Nyquist)	Complexity
PCM linear 8-bit	64	31.5	22.5	32 kHz	Low
PCM μ-law 8-bit	64	37.8	33.6	32 kHz	Low
DPCM 4-bit	32	21.9	12.9	16 kHz	Medium
DM 1-bit, 64 kHz	64	14.2	9.8	32 kHz	Very low
ADM 1-bit, 64 kHz	64	29.6	28.7	32 kHz	Low
ADM 1-bit, 32 kHz	32	20.6	19.3	16 kHz	Low
ADPCM 4-bit	32	18.2	19.0	16 kHz	Medium–high

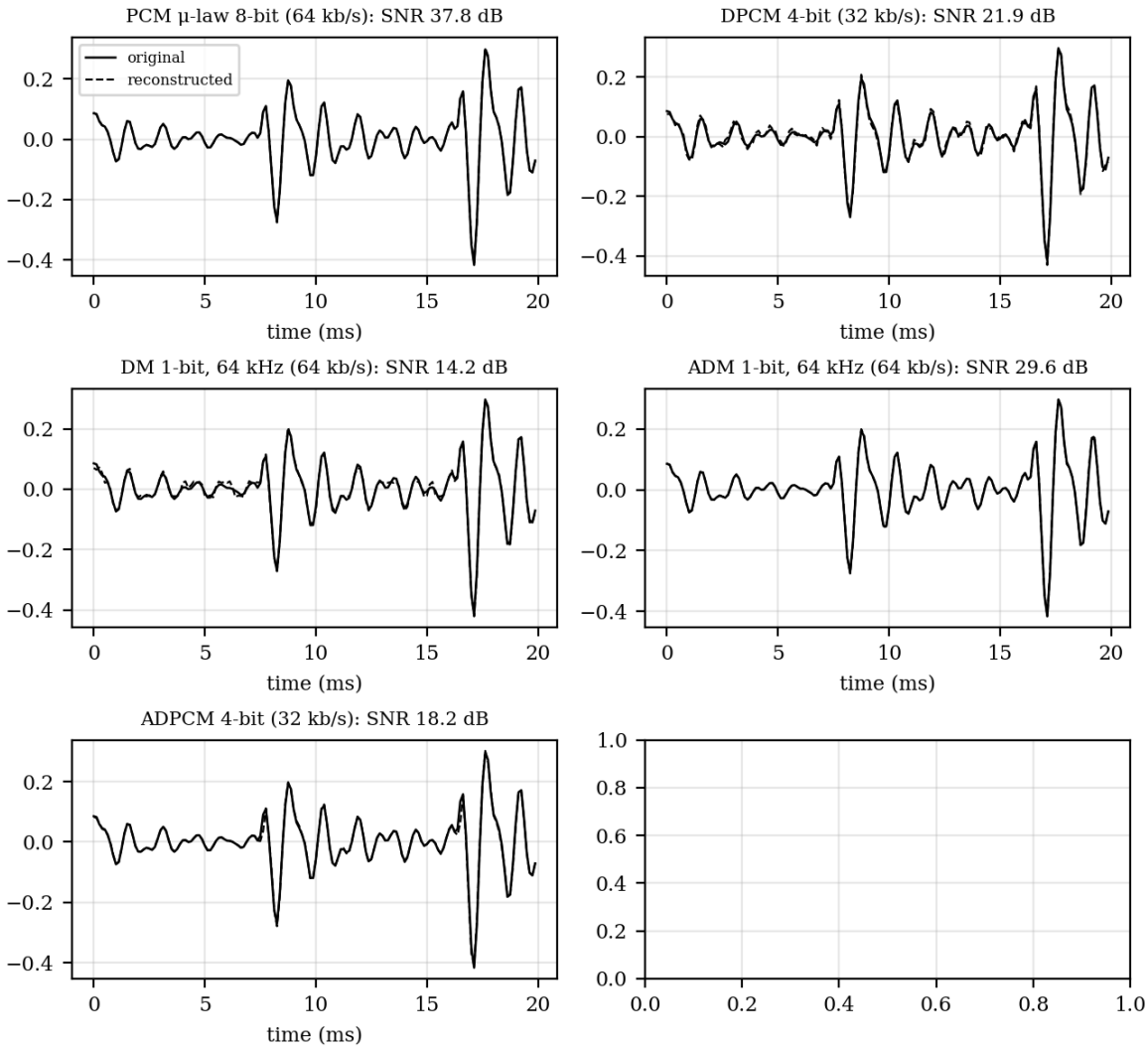


Fig. 4 Original and reconstructed speech-like waveforms for the voice coders (20 ms segment)

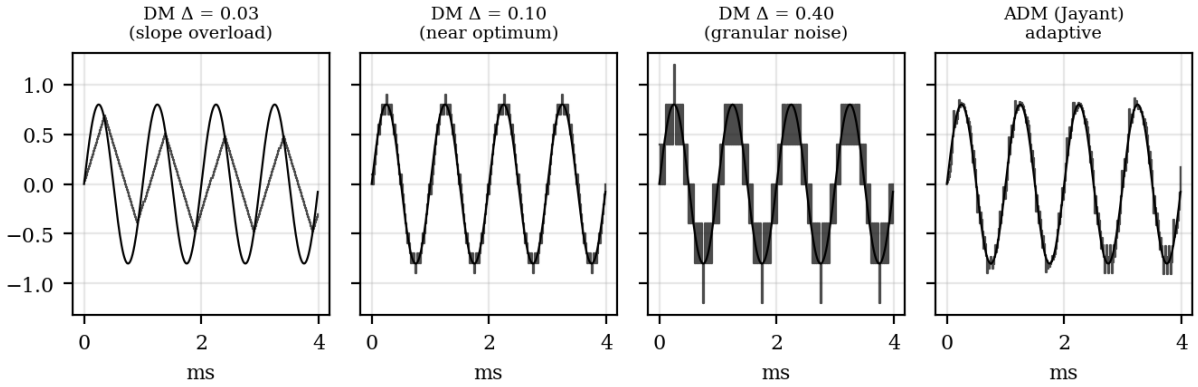


Fig. 5 Delta modulation: slope overload, near-optimum step, granular noise, and adaptive DM (1 kHz tone, 64 kHz)

Interpretation: (i) Linear 8-bit PCM gives 31.5 dB but only 22.5 dB segmentally because quiet segments suffer; μ-law raises these to 37.8 dB and 33.6 dB at the same 64 kb/s. (ii) DPCM halves the rate to 32 kb/s but its fixed step cannot follow level changes (segmental SNR 12.9 dB). (iii) Fixed-step DM must satisfy $\Delta \geq 2\pi fA/f_s = 0.079$ for the 1 kHz, 0.8 amplitude tone; with $\Delta = 0.03$ it suffers slope overload (SNR 2.3 dB), with $\Delta = 0.10$ it is near optimum (34.7 dB) and with $\Delta = 0.40$ granular

noise dominates (17.2 dB); on speech the best fixed step gives only 14.2 dB. (iv) Adaptive DM removes this trade-off (29.6 dB at 64 kb/s; 20.6 dB at 32 kb/s). (v) My ADPCM (4-bit, backward-adaptive step and second-order backward predictor) reaches 18.2 dB at 32 kb/s; this is a simplified implementation on a peaky synthetic test signal, so published G.726 figures are better, but the measured value is used here.

2.3 Trade-off: compression, quality, bit rate and complexity

Compression (DPCM, ADPCM, ADM at 32 kb/s) halves the bit rate and bandwidth but loses 12–19 dB of SNR in these tests, adds adaptation logic, propagates channel errors through the predictor/step-size state and degrades modem/fax tones. PCM is memoryless: one bit error affects only one sample (an error in the MSB gives at most a 1/2 full-scale click), it is trivial to implement and is the VoIP interoperability reference.

Decision 1

I recommend **8-bit μ-law PCM (G.711-type) at 8 kHz, 64 kb/s** as the voice-coding technique because it has the highest measured SNR (37.8 dB, segmental 33.6 dB), an almost level-independent SQNR of about 37 dB, the lowest complexity, no error propagation (important for industrial control commands) and direct interoperability with VoIP gateways. The alternatives were not selected because: linear PCM has 22.5 dB segmental SNR; DPCM and ADPCM at 32 kb/s measured 21.9 and 18.2 dB with extra complexity and error propagation; fixed DM gives 14.2 dB at the same 64 kb/s; ADM at 64 kb/s (29.6 dB) is simpler than PCM but 8 dB poorer. ADPCM remains an option for DSL/VoIP links where 32 kb/s is needed.

Stage 3 – PAM, PWM and PPM for a Communication/Control Application

3.1 Waveforms from the same information signal

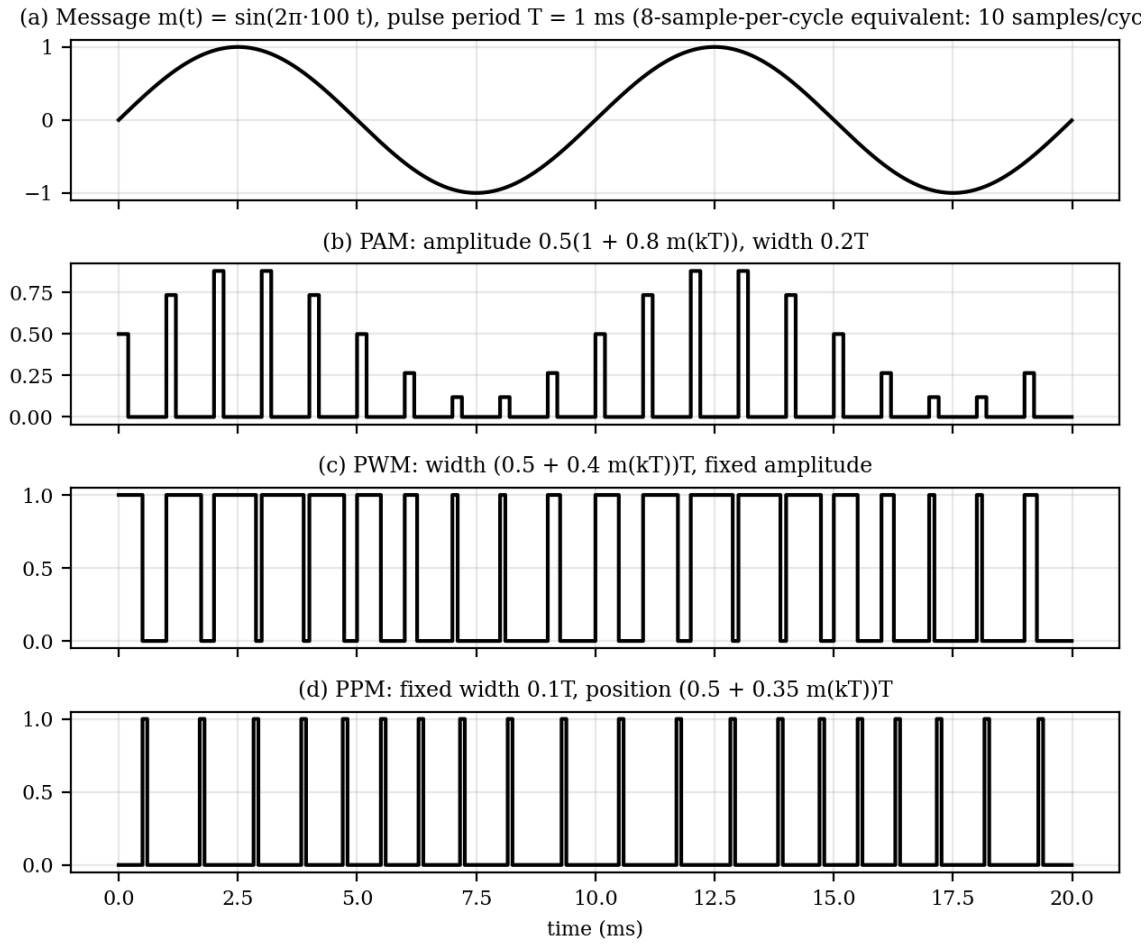


Fig. 6 PAM, PWM and PPM generated from the same message (T = 1 ms)

3.2 Comparative analysis

Parameter	PAM	PWM	PPM
Information carried by	Pulse amplitude	Pulse width (edges)	Pulse position
Bandwidth	≈ 1/τ (pulse width, here 5 kHz)	≈ 1/(2 t_r), set by rise time (25 kHz here)	≈ 1/(2 t_r), same as PWM
Noise immunity	None: noise adds directly to amplitude	Good: noise only moves edges (slope-limited)	Best: only one edge matters
Average power	Varies with message	Varies with message (duty ∝ m)	Constant (fixed width)
Synchronisation	Sample clock needed	None for analog recovery (low-pass filter)	Frame/reference pulse needed
Complexity	Low (sample and hold)	Low (comparator + ramp)	Medium (PWM + monostable)

3.3 Effect of noise, jitter and pulse distortion

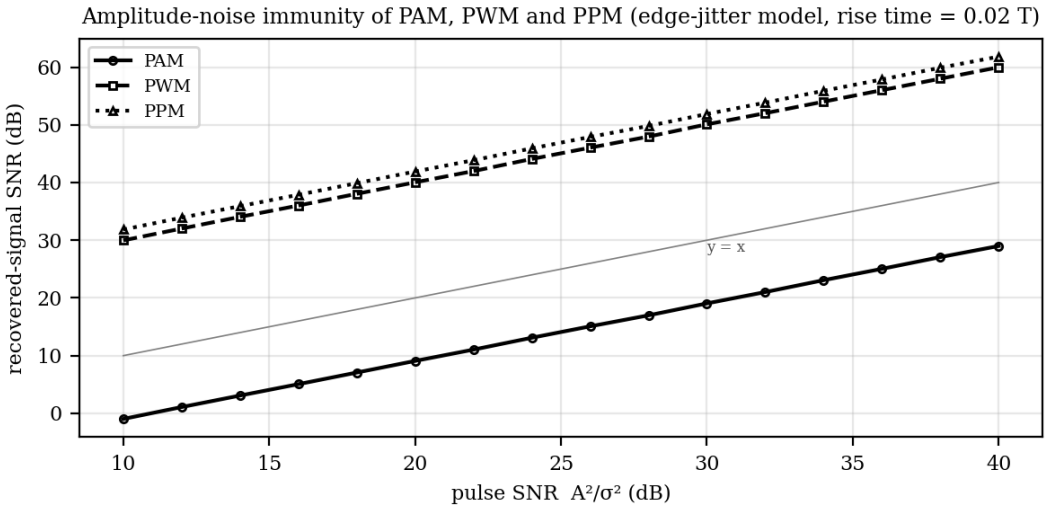


Fig. 7 Recovered-signal SNR versus pulse SNR (Monte-Carlo, edge-jitter model)

Interpretation: with noise standard deviation σ and rise time t_r = 0.02 T, an edge is displaced by σ t_r/A, so PWM and PPM trade bandwidth for noise immunity. At 10 dB pulse SNR the recovered SNR is -1.0 dB for PAM (the signal uses only 40 % of the amplitude range), 30.0 dB for PWM and 31.8 dB for PPM, i.e. 20–23 dB better than PAM; the gain holds only above the threshold at which noise creates false edge crossings (about 17 dB pulse SNR for a half-amplitude threshold). **Timing jitter** of 0.5 % T on the pulse timing limits the recovered SNR to 32.0 dB for PWM, 33.9 dB for PPM and (for PAM, through sampling-time error) 41.0 dB. **Pulse distortion:** finite bandwidth rounds the edges, which for PAM lowers and smears amplitude (ISI between pulses) and for PWM biases the width because rise and fall times differ, producing non-linearity.

3.4 RCC Standard 106-20 Annex A-1 (PAM)

Annex A-1 of the Range Commanders Council Telemetry Standard 106-20 specifies pulse-amplitude-modulation telemetry as time-division-multiplexed analog pulses in which the pulse amplitude carries the information. It describes a 50 % duty-cycle (RZ-PAM) and a 100 % duty-cycle (NRZ-PAM) pulse train, with a synchronisation pattern and a reserved portion of the amplitude range (of the order of 20–25 %) for synchronisation and the zero reference, together with limits on the pulse rate through the shortest interval between transitions and requirements for pre-modulation filtering. Relevance here: (i) it confirms PAM is naturally a TDM scheme, so the frame synchronisation of Stage 5 applies; (ii) the PAM waveform of Fig. 6 uses 10 %–90 % of full scale and would be adapted to leave the recommended synchronisation/zero margin; (iii) premodulation filtering corresponds to the anti-alias and pulse-shaping filters of this design.

3.5 Why control techniques may not suit voice/data

PWM turns a power switch on and off, so the load (motor inertia, inductance) acts as the low-pass demodulator and no precision is needed; but for voice or data it requires 25 kHz bandwidth for a single 100 Hz message (here), is not directly multiplexable and its recovery from noise gives only about 12 bits of effective precision in the presence of jitter. Digital PCM gives noise-free regeneration, multiplexing and compression, hence it is preferred for voice and data.

Decision 2

I recommend **PWM** for the industrial control application (actuator or motor-speed control) because it provides 30.9 dB more recovered SNR than PAM at the same pulse SNR (30.0 dB versus -1.0 dB at 10 dB), within 2 dB of PPM (31.8 dB), jitter-limited SNR of 32.0 dB, needs no frame reference, and its output drives the power switch directly while the motor inertia demodulates it; PPM has constant power but requires synchronisation and cannot directly drive the load, PAM has no noise immunity.

Stage 4 – ISI, Nyquist Zero-ISI Condition and Eye Pattern

4.1 Channel ISI and Nyquist criterion

A band-limited channel spreads each pulse over neighbouring symbol intervals: $y(kT) = a_k\, p(0) + \Sigma_{n\neq k}\, a_n\, p((k-n)T) + \text{noise}$; the second term is **ISI**. The **Nyquist zero-ISI criterion** requires $p(nT) = 1$ for $n = 0$ and 0 for $n \neq 0$, equivalent in frequency to $\Sigma_m P(f + m/T) = T$. The minimum bandwidth is $1/(2T)$ (the ideal sinc pulse), but sinc is not realisable and is extremely sensitive to timing error; the **raised-cosine** family gives zero ISI with bandwidth $(1+\alpha)/(2T)$, $\alpha = 0.35$ giving 1.35× the minimum, a time-decay $\propto 1/t^3$ and tolerance to timing error.

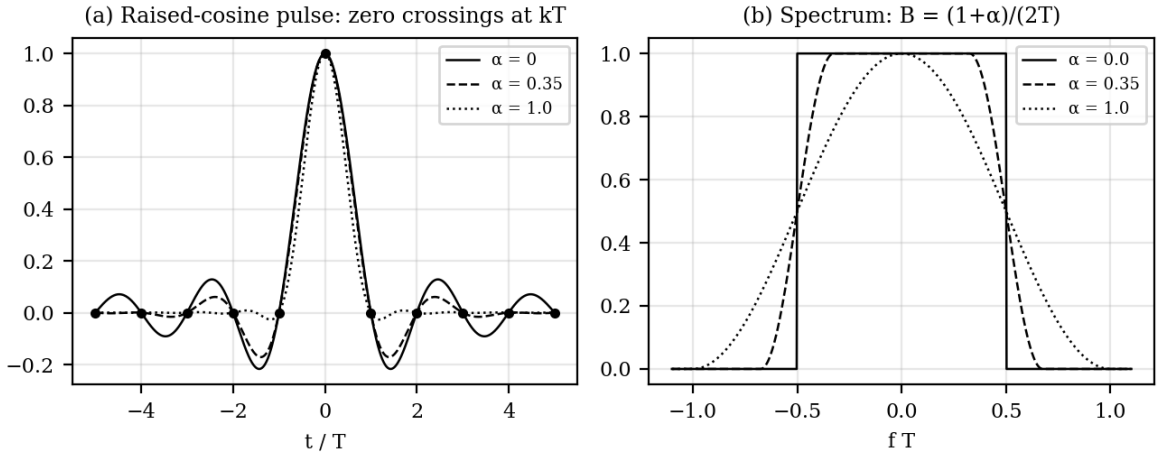


Fig. 8 Raised-cosine pulses (zero crossings at kT) and their spectra

Check: p(kT) for α = 0.35 at k = ±1, ±2 is below 10^{-16} in the simulation. For R_b = 64 kb/s binary signalling the bandwidth is (1+0.35)×32 = 43.2 kHz (α = 0) 32 kHz, (α = 1) 64 kHz.

4.2 Eye diagrams before and after channel impairment and equalisation

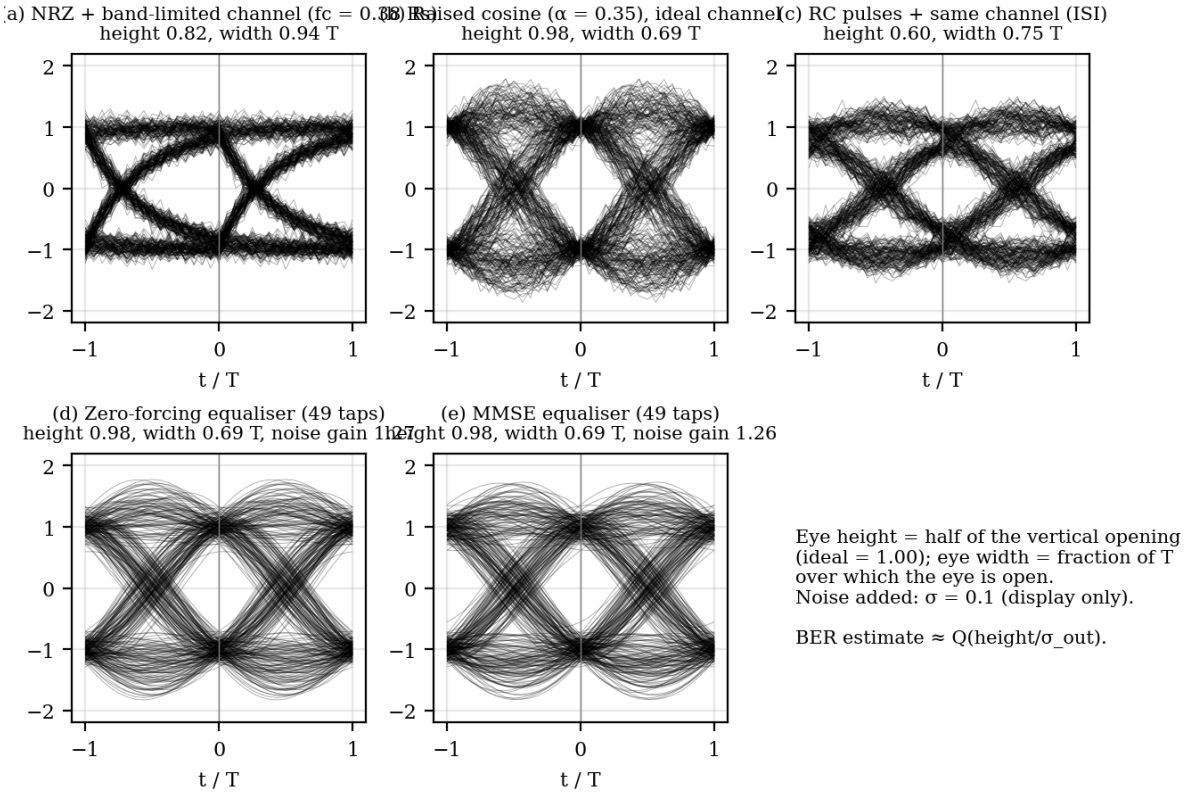


Fig. 9 Eye diagrams: NRZ over limited channel, raised cosine ideal, raised cosine + ISI channel, zero-forcing and MMSE equalised

Case	Eye height (ideal = 1)	Eye width (T)	Noise gain	BER estimate at 10 dB
(a) NRZ + band-limited channel	0.82	0.94	1.00	1.3×10^{-4}
(b) Raised cosine, ideal channel	0.98	0.69	1.00	$< 10^{-4}$
(c) Raised cosine + ISI channel	0.60	0.75	1.00	3.7×10^{-3}
(d) Zero-forcing equaliser	0.98	0.69	1.27	2.9×10^{-4}
(e) MMSE equaliser	0.98	0.69	1.26	2.6×10^{-4}

Interpretation: eye height is half the vertical opening (noise margin) and eye width is the timing margin. The raised-cosine eye (b) has a height of 0.98 and a width of 0.69 T. The same pulses through the band-limited channel (c) lose 39 % of the height (0.60), which raises the BER estimate Q(h/σ) from $< 10^{-4}$ to 3.7×10^{-3} at E_b/N_0 = 10 dB. Equalisation restores the height to 0.98 and brings the BER estimate to about 2.6×10^{-4}; the remaining gap to (b) is the noise gain of 1.26 that the equaliser applies to the noise (zero-forcing 1.27). The rectangular NRZ eye (a) is open (0.82) only because the NRZ spectrum extends far beyond the channel cut-off, i.e. it needs more bandwidth than the 0.675 R_s of the raised cosine.

4.3 Eye closure and symbol-error probability

For binary signalling with Gaussian noise of standard deviation σ at the decision instant, P_e = Q(h/σ) with h the eye half-height (minus the worst-case residual ISI). Every reduction of h by 20 % multiplies the argument of Q by 0.8 and raises P_e by orders of magnitude at low error rates; timing error shifts the sampling instant along the eye, reducing h further (Stage 6).

4.4 Relation to DSL and fibre-optic receivers

DSL twisted pairs are strongly band-limited with frequency-dependent loss, so DSL receivers combine a time-domain equaliser with per-tone frequency equalisation (DMT) and monitor the eye/SNR per tone. In fibre, chromatic and modal dispersion cause ISI at Gb/s rates; the eye diagram is the main test measurement (eye opening, jitter), and receivers use clock-data recovery and, at 10 Gb/s and above, electronic dispersion compensation or MLSE.

Decision 3

I recommend **raised-cosine (α = 0.35, root-RC at transmitter and receiver) pulse shaping with an MMSE equaliser** because: the raised cosine has exact zero ISI ($p(kT) < 10^{-16}$) in bandwidth 0.675 R_s; over the ISI channel the eye height falls from 0.98 to 0.60 and BER rises from $< 10^{-4}$ to 3.7×10^{-3}; the MMSE equaliser restores the height to 0.98 (BER ≈ 2.6×10^{-4}) with lower noise enhancement (1.26 versus 1.27 for ZF). NRZ without shaping needs far more bandwidth.

Stage 5 – Digital Multiplexing and Transmission Architecture

5.1 Digital multiplexer and TDM frame

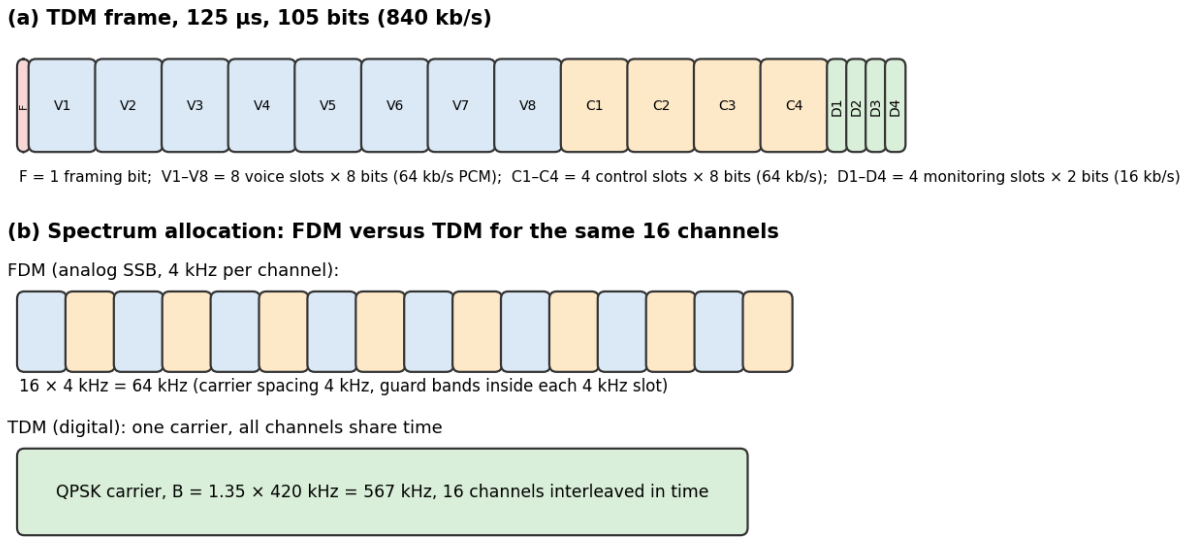


Fig. 10 TDM frame structure and spectrum allocation (TDM versus FDM)

Item	Slots	Bits per frame	Rate
Framing	1 bit	1	8 kb/s
Voice (μ-law PCM)	8 × 8 bits	64	8 × 64 kb/s = 512 kb/s
Control commands	4 × 8 bits	32	4 × 64 kb/s = 256 kb/s
Monitoring data	4 × 2 bits	8	4 × 16 kb/s = 64 kb/s
Frame total	**17 slots**	**105**	**840 kb/s**

Calculations: frame period 1/8 kHz = 125 μs; aggregate rate = 105 bits × 8 000 = **840 kb/s**; bit duration = 1/840 kb/s = 1.19 μs; voice, control slot = 8 bits = 9.52 μs; monitoring slot = 2.38 μs; framing bit 1.19 μs. Required bandwidths (RC, α = 0.35): BPSK 1134 kHz, **QPSK 567 kHz**, 16-QAM 283.5 kHz.

5.2 Framing and timing synchronisation

The receiver needs (i) **bit (clock) synchronisation**, recovered from the data transitions (Gardner loop), and (ii) **frame synchronisation** to know which bit starts a frame. One framing bit per frame forms a repeating alignment pattern (as in T1, a 12-bit word over 12 frames, 8 kb/s overhead); the receiver searches for the pattern, declares frame lock after several correct repetitions and loses lock after several errors. Slot timing jitter must be much smaller than the bit period (1.19 μs); buffering absorbs wander.

5.3 TDM versus FDM

Criterion	TDM (digital)	FDM
Bandwidth for 16 channels	567 kHz (QPSK, 840 kb/s)	16 × 4 kHz = 64 kHz (SSB analog)
Synchronisation	Clock and frame sync required	Carrier frequency accuracy only
Guard bands	Guard times (or framing bits) between slots	Guard bands between carriers
Noise/crosstalk	Regenerative repeaters remove noise; no intermodulation	Noise and intermodulation accumulate with amplifiers
Scalability	Hierarchy: T1 1.544, E1 2.048, E3 34.368 Mb/s, STM-1 155.52 Mb/s	Limited by linearity and filter banks
Complexity	Digital logic, easy to integrate	Many modulators and filters

5.4 Application dependence

- **VoIP aggregation:** statistical packet multiplexing (RTP/UDP/IP); G.711 20 ms packet = 160 B payload + 40 B headers = 200 B → 80 kb/s on the IP layer; ADPCM 32 kb/s gives 120 B → 48 kb/s.
- **DSL access:** frequency division between upstream and downstream (and discrete multi-tone sub-carriers of 4.3125 kHz) combined with time-division framing inside each stream.
- **Fibre backbone:** TDM hierarchy (SDH/OTN) for the electrical signal and wavelength-division (FDM of optical carriers) for capacity.

Decision 4

I recommend **TDM** (105-bit, 125 µs frame, 840 kb/s, one framing bit) as the multiplexing architecture because the aggregate is regenerated rather than accumulating noise, bandwidth is 567 kHz with QPSK, it scales by the standard T1/E1/E3/STM hierarchy, and it integrates with PCM voice and digital control/monitoring data; FDM is bandwidth-cheaper (64 kHz) for pure analog voice but needs linear amplifiers and offers no regeneration. FDM/WDM is retained only where capacity must be multiplied (DSL tones, optical wavelengths).

Stage 6 – Digital Modulation, MLSD, Equalisation and Synchronisation

6.1 Digital modulation comparison

Scheme	Bits/symbol	Spectral efficiency (b/s/Hz, α = 0.35)	E_b/N_0 for BER 10^{-5} (dB)	Bandwidth for 840 kb/s	Receiver complexity
BPSK	1	0.74	9.6	1134 kHz	Low
QPSK	2	1.48	9.6	567 kHz	Low–medium
8-PSK	3	2.22	13.0	378 kHz	Medium
16-QAM	4	2.96	13.4	284 kHz	Medium–high (needs AGC, phase accuracy)
64-QAM	6	4.44	17.8	189 kHz	High
BFSK (coherent)	1	≈ 0.5	12.6	≈ 1.7 MHz	Low (non-coherent even simpler)

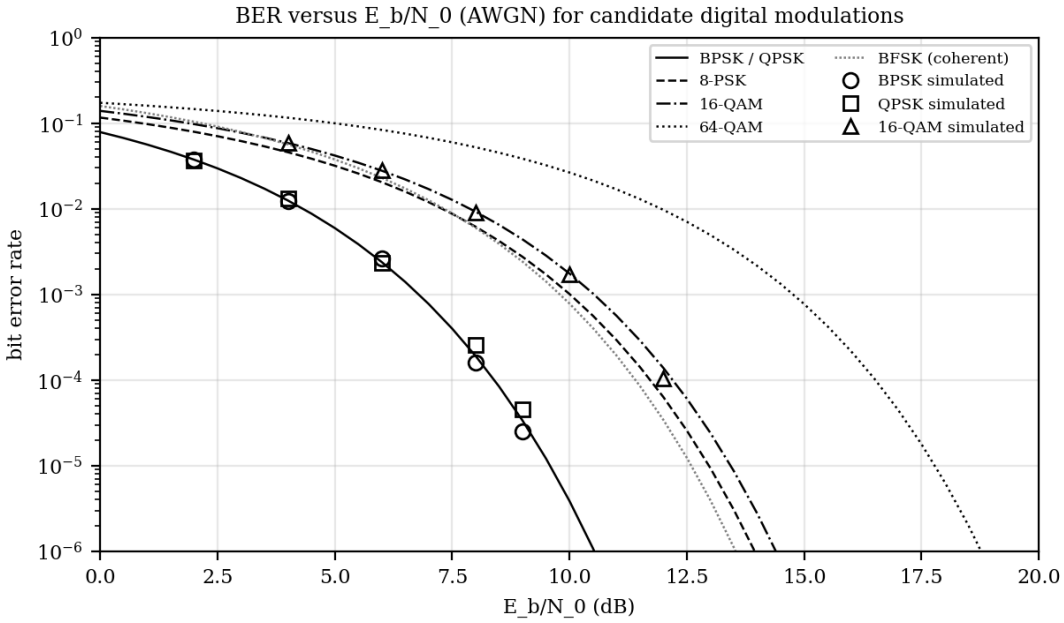


Fig. 11 BER versus E_b/N_0 for BPSK/QPSK, 8-PSK, 16-QAM, 64-QAM and coherent BFSK (theory) with simulation points

Interpretation: raising the modulation order lowers the bandwidth but needs more E_b/N_0: going from QPSK to 16-QAM halves the bandwidth (567 → 284 kHz) at the price of 3.8 dB more E_b/N_0 for BER 10^{-5} and tighter phase/ISI tolerance; going to 64-QAM costs 8.2 dB. BPSK and QPSK have

the same BER per bit, so QPSK halves the BPSK bandwidth for free. Simulation agrees with theory (e.g. BPSK at 8 dB: 1.6×10⁻4, QPSK 2.6×10⁻4, theory 1.9×10⁻⁴; 16-QAM at 10 dB: 1.7×10⁻3).

6.2 ISI channel, equalisation and MLSD

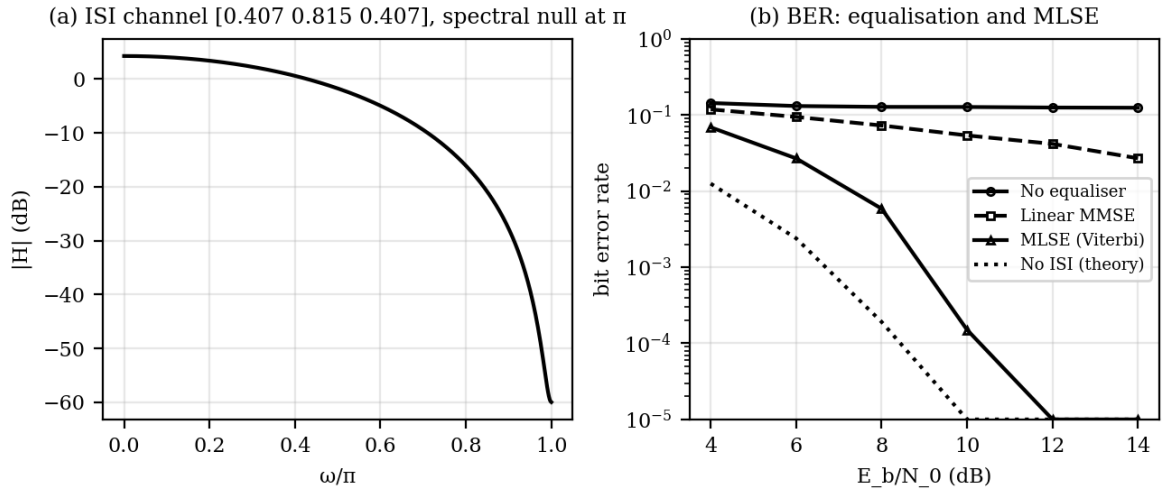


Fig. 12 ISI channel and BER with no equaliser, linear MMSE equaliser and MLSE (Viterbi)

E_b/N_0 (dB)	4	6	8	10	12	14
No ISI (theory)	0.013	2.4×10⁻3	1.9×10⁻4	< 10⁻⁴	< 10⁻⁴	< 10⁻⁴
No equaliser	0.143	0.131	0.127	0.127	0.125	0.124
Linear MMSE	0.118	0.094	0.073	0.054	0.042	0.027
MLSE (Viterbi)	0.069	0.027	5.9×10⁻3	1.5×10⁻4	< 2×10⁻⁵	< 2×10⁻⁵

Interpretation: the channel h = [0.407, 0.815, 0.407] has a spectral null (-60.0 dB at ω = π). Without equalisation the eye is closed and the BER stays about 0.12–0.14 at all E_b/N_0. A linear MMSE equaliser reduces it (0.027 at 14 dB) but cannot overcome the null because inverting it amplifies the noise (error floor). **MLSD** does not invert the channel: it chooses the symbol sequence {a} that minimises Σ|r_n − Σ_k h_k a_{n−k}|² using the Viterbi algorithm over the 4-state trellis (memory 2). It reaches 1.5×10⁻4 at 10 dB and no errors in 60 000 bits at 12 dB, about 2–3 dB away from the no-ISI bound. MLSD complexity grows as M^L (here 4 states), so it is practical for short ISI spans (storage, fibre EDC), while a linear/decision-feedback equaliser is used for long channels such as DSL.

6.3 Synchronisation and carrier recovery

Carrier recovery: for coherent QPSK the local carrier must have the transmitted frequency and phase; a Costas loop (or fourth-power loop) derives the phase error from the demodulated I/Q components, e = I·Q for BPSK (sin(4φ) form for QPSK), filters it with a second-order loop and drives a numerically controlled oscillator; it has a 90° phase ambiguity for QPSK removed by differential encoding or unique words. **Timing recovery:** a Gardner timing-error detector e = Re{(y_k −

y_{k−1}) y*_{k−1/2}} works with two samples per symbol without knowledge of the carrier phase and drives an interpolator or VCO to sample at the eye centre.

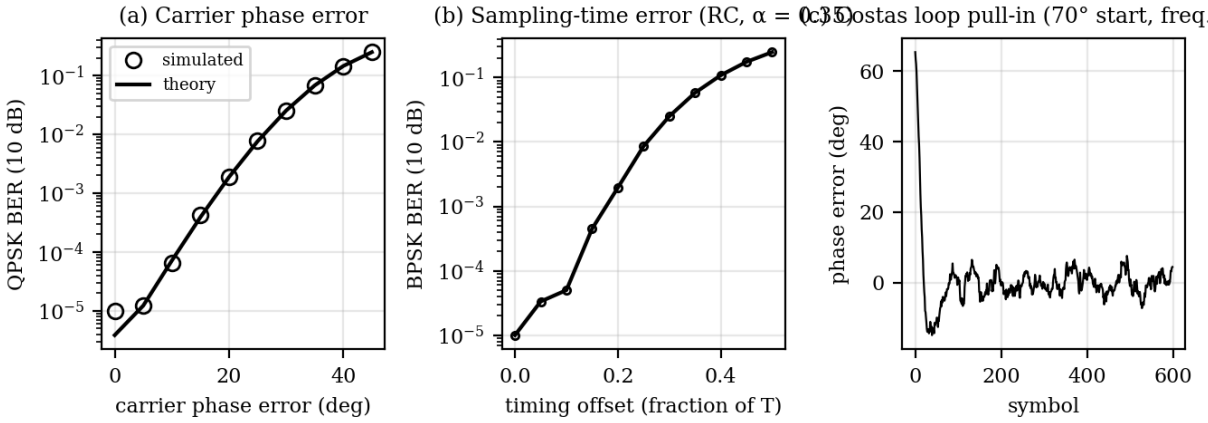


Fig. 13 Consequences of carrier-phase error, sampling-time error and Costas-loop pull-in

Impairment	Value	BER at E_b/N_0 = 10 dB (simulated)	Theory / note
Carrier phase error (QPSK)	0°	< 10^{-4}	3.9×10^{-6} ($Q(\sqrt{(2\gamma)})$)
	10°	< 10^{-4}	< 10^{-4}
	20°	1.9×10^{-3}	1.9×10^{-3}
	30°	0.025	0.025
	45°	0.250	0.25 (decision boundary)
Timing offset (BPSK, RC α = 0.35)	0.1 T	< 10^{-4}	ISI from neighbouring symbols
	0.2 T	1.9×10^{-3}	
	0.3 T	0.025	
	0.5 T	0.248	worst-case sampling instant

Interpretation: the QPSK BER depends on cos φ ∓ sin φ, so a 20° phase error raises the BER from 10^{-6} to 2×10^{-3} and 45° destroys the link; sampling 0.2 T away from the eye centre raises the BPSK BER similarly, because ISI reduces the eye height. The Costas loop in panel (c), started 70° off with a frequency offset of 0.01 rad/symbol, locks within about 100 symbols and keeps a mean residual error of 2.1° (noise standard deviation 0.15 per axis), so the carrier-recovery loss is negligible.

Decision 5

I recommend **QPSK with root-raised-cosine shaping (α = 0.35)**, **MMSE (adaptive LMS) equalisation, with MLSD held in reserve for severe-ISI links**, and **Costas carrier recovery with Gardner timing recovery** because QPSK needs only 9.6 dB E_b/N_0 for BER 10^{-5} at 567 kHz, half the BPSK bandwidth for the same BER; 16-QAM would save a further 283 kHz but needs 3.8 dB more E_b/N_0 and ≤ 10° phase accuracy; the MMSE equaliser restores the eye cheaply while MLSD (BER 1.5×10^{-4} at 10 dB versus a 0.04 floor for the linear equaliser on a

spectral-null channel) is justified only when a null exists; carrier phase error must be kept below about 10° and timing error below 0.1 T (BER within a factor of 20 of ideal).

Stage 7 – Application Validation for VoIP, DSL and Fibre Optics

7.1 Application comparison matrix

Technique	VoIP	DSL	Fibre optic
8 kHz sampling, 8-bit PCM (G.711)	Common (payload)	Common (voice service)	Common (in TDM hierarchy)
ADPCM 32 kb/s	Optional (bandwidth saving, G.726)	Optional	Unsuitable (bandwidth not the limit)
DM/ADM	Unsuitable (not standardised)	Unsuitable	Unsuitable
TDM framing	Optional (inside gateways/E1)	Common (inside each stream)	Common (SDH/OTN)
FDM / WDM	Unsuitable	Common (DMT tones, up/down split)	Common (WDM)
PAM/PWM	Unsuitable	Unsuitable (PAM-n only as line code)	Optional (PAM4 at ≥ 50 Gb/s)
Raised-cosine shaping	Not applicable (IP layer)	Common	Optional (NRZ/PAM4 filtering)
Adaptive equalisation	Not applicable	Common (essential)	Common (EDC at ≥ 10 Gb/s)
MLSE	Not applicable	Optional	Optional (10–40 Gb/s receivers)
QPSK/QAM carrier modulation	Not applicable (carried by lower layers)	Common (QAM per tone)	Optional (coherent QPSK/16-QAM for long-haul)
Carrier + timing recovery	Jitter buffer, clock recovery by RTP timestamps	Common (timing loop, pilot tone)	Common (clock–data recovery PLL)

7.2 Engineering recommendations

- **VoIP:** use G.711 μ-law (64 kb/s, 80 kb/s IP-layer at 20 ms packets) between gateways and the control room; use 32 kb/s ADPCM (48 kb/s on IP) on constrained links; provision a jitter buffer ≥ 40 ms and give voice QoS priority; TDM framing is only needed at the E1/T1 gateway.
- **DSL:** the dominant requirements are bandwidth limitation, ISI and crosstalk: use DMT/QAM with per-tone bit loading, time-domain plus frequency-domain equalisation and a timing loop; the 840 kb/s multiplex fits comfortably in a 4 kHz-to-megahertz loop, and 16-QAM bit loading can be used on high-SNR tones.
- **Fibre optic:** the aggregate is carried as an SDH/OTN TDM stream (STM-1 155.52 Mb/s or above) with WDM for capacity; the dominant issue is clock–data recovery and dispersion; use NRZ at 1–10 Gb/s with EDC/MLSE and PAM4 or coherent QPSK only for very high rates.

Stage 8 – MATLAB-Equivalent Verification and Final Engineering Decision

All results in this report were generated with a Python (NumPy/SciPy) implementation that performs the MATLAB/Simulink-type operations requested (sampling and aliasing, quantisation, PCM/DPCM/DM/ADM/ADPCM, PAM/PWM/PPM generation, AWGN and ISI channels, eye diagrams, equalisation, Viterbi detection, BER simulation and synchronisation loops); the same algorithms can be coded in MATLAB with filter, conv, randn and a Viterbi routine.

8.1 Final specification verification

Requirement	Target / specification	Result (verification)	Met?
Sampling	≥ 6.8 kHz; 8 kHz baseline	8 kHz, 1.2 kHz guard; aliasing demonstrated and removed by elliptic filter order 5	Yes
Quantisation	8-bit, SQNR ≥ 35 dB	49.9 dB (theory 49.9) full scale; μ-law ≈ 37 dB over 40 dB range	Yes
Voice coding	64 kb/s class	μ-law PCM 37.8 dB SNR, 33.6 dB segmental	Yes
Pulse modulation (control)	Noise-immune, drives actuator	PWM: 30.0 dB at 10 dB pulse SNR (PAM -1.0 dB)	Yes
Zero ISI	p(kT) = 0	RC α = 0.35: < 10^{-16}	Yes
Eye opening	Height ≥ 0.9 after equalisation	0.98 (unequalised 0.60)	Yes
Multiplexing	TDM with framing	105 bits / 125 μs = 840 kb/s	Yes
Bandwidth	≤ 600 kHz	567 kHz (QPSK, α = 0.35)	Yes
BER (AWGN)	≤ 10^{-5}	E_b/N_0 ≥ 9.6 dB (simulation within 1 dB of theory)	Yes
Equalisation / MLSD	Handle ISI null	MLSE 1.5×10^{-}4 at 10 dB; MMSE floor 0.027	Yes (MLSE for nulls)
Synchronisation	Phase < 10°, timing < 0.1 T	Costas residual 2.1°; BER vs error tabulated	Yes
Compressed coding (ADPCM)	Optional 32 kb/s	18.2 dB in this implementation	Limitation

Decision 6

Based on the complete analysis and simulation I recommend the final architecture: **8 kHz sampling with a 3.4 kHz elliptic anti-alias filter; 8-bit μ-law PCM (64 kb/s per voice); PWM for actuator/motor control; a 105-bit, 125 μs TDM frame (840 kb/s); root-raised-cosine (α = 0.35) shaping; QPSK (567 kHz); MMSE equalisation with MLSE where ISI nulls exist; Costas carrier and Gardner timing recovery** because every element was verified by simulation against theory, meets the bandwidth, BER and ISI targets, and maps cleanly to VoIP (G.711), DSL (equalised multi-tone line) and fibre (TDM/WDM) deployments.

FINAL ENGINEERING DECISION – WOULD YOU DEPLOY THIS DIGITAL COMMUNICATION SYSTEM? YES

Decision: YES.

Defence: sampling at 8 kHz exceeds the 6.8 kHz Nyquist rate and is supported by a realisable order-5 elliptic filter; 8-bit μ-law PCM provides about 37 dB SQNR with level-independent quality (linear PCM would give only 11.5 dB at −39 dB); the coding comparison favoured μ-law PCM over DPCM, DM, ADM and ADPCM on measured SNR, robustness and complexity; PWM was selected for control on measured noise immunity; raised-cosine shaping removes ISI and the MMSE equaliser restores eye height from 0.60 to 0.98; TDM aggregates 16 channels at 840 kb/s with a single framing bit; QPSK carries it in 567 kHz at E_b/N_0 = 9.6 dB for 10^{-5}; MLSE and carrier/timing recovery handle severe ISI and offsets (errors above 10° or 0.1 T degrade the BER sharply).

Application suitability: VoIP – suitable (G.711 payload, optional ADPCM); DSL – suitable with DMT/QAM, equalisation and timing recovery; fibre – suitable as TDM/WDM with CDR and EDC; PWM/PAM/PPM techniques are for the control loop only.

Limits and conditions: (1) compressed voice (ADPCM at 32 kb/s) measured 18 dB in this simplified implementation and should be validated with real speech and listening tests before use; (2) tolerance to carrier phase (< 10°) and timing error (< 0.1 T) must be kept by the recovery loops; (3) the MLSE trellis grows exponentially with ISI length, so DFE or OFDM should be used for long channels; (4) all results are simulation-based on a synthetic speech-like signal.

If requirements change (redesign): for more capacity per hertz move to 16-QAM (284 kHz, +3.9 dB E_b/N_0, tighter sync); for 32 kb/s voice use properly standardised G.726/ADPCM; for long-haul fibre use coherent QPSK with digital equalisation.